

MSE 223 Introduction to Fluid Mechanics - Final Project

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- Using past data
- Using data from others
- Using any text from past reports or other student's reports

Generative AI declaration Statement: During the preparation of this work the author used ChatGPT as a search engine in order to source external background and information about the topics such as assessing the system to see if atmospheric pressure can be neglected. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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Introduction:

This project explores the application of fluid mechanics principles to the design and analysis of a small scale real world hydroelectric dam intended for renewable energy generation in Peace River, British Columbia. Many remote communities in Canada continue to rely on diesel-powered electricity, which is both costly and carbon-intensive. As an Engineer in training for BC Hydro, the given task is to evaluate the technical feasibility and performance of a proposed dam system, which includes a reservoir, penstock, and turbine system. The goal is to optimize the efficiency of energy production while taking into account environmental sustainability and system losses. By examining key parameters such as hydrostatic pressure, flow velocity, head loss, and power output, the analysis aims to determine whether the proposed system could offer a viable renewable energy solution. The analysis not only determines the mechanical and energy output but also highlights the broader environmental implications and potential design improvements that can enhance system performance. This report outlines the methodology, calculations, and insights derived from fluid mechanics theory to support sustainable hydro energy development.

Design Analysis

1. Hydroelectric dams present a number of technical and environmental advantages, making them one of the most widely used forms of renewable energy. They offer a reliable and consistent power supply and can respond quickly to fluctuations in demand. Once constructed, hydroelectric systems have relatively low operating and maintenance costs and a long service life. They produce no greenhouse gas emissions during operation, contributing significantly to climate change mitigation. Additionally, reservoirs created by dams can be used for water storage, irrigation, and recreation.

Some of the demerits include the initial capital cost for construction is very high, and the process of damming a river can have significant ecological consequences. Dams disrupt natural water flow, block fish migration, alter sediment transport, and can degrade the downstream water quality. They also have the potential to displace communities and wildlife due to flooding of large areas. In some cases, decomposing vegetation in reservoirs can lead to the release of methane, a potent greenhouse gas. Therefore, while hydroelectricity is a powerful and clean energy source, it must be carefully planned and managed to minimize environmental harm.

2.-5. Refer to Figure 1 in Appendix

6.-10. Refer to Figure 2 in Appendix

11. The flow characteristics within a hydroelectric system play a crucial role in determining its overall efficiency. The Reynolds number for flow in the penstock is 25,409, which indicates turbulent flow. Turbulent flow is typical in large-scale hydro systems due to the high flow rates and pipe diameters involved. While turbulent flow results in greater frictional losses compared to laminar flow, which reduces the effective head available to the turbine, it also has practical benefits. Turbulent flow can help prevent sedimentation and promotes better mixing, which can reduce maintenance requirements and improve long-term system reliability. In large-scale applications, achieving laminar flow is impractical as it would require extremely low velocities and thus very low energy production. Therefore, though turbulent flow is less efficient from a theoretical standpoint, it is the standard in hydroelectric systems and optimized by design to maintain high performance.

12. Minimizing frictional head losses in the penstock is important for maximizing the efficiency of a hydroelectric power system. A possible strategy is to use materials with low surface roughness like smooth steel or polymer linings for the inner walls of the penstock, which reduces the friction. Regular maintenance is also important to prevent buildup of biofilms, scaling, or corrosion, all of which can increase roughness and losses. Designing the penstock with gradual bends and minimizing abrupt changes in direction/geometry helps to reduce local turbulence and energy dissipation. Additionally, increasing the diameter of the penstock decreases the velocity for a given flow rate, which in turn can reduce frictional losses. Optimizing flow conditions at the inlet can further reduce disturbances that might contribute to head loss. By combining material choices, design, and proper maintenance, the efficiency of hydroelectric systems can be greatly enhanced through the reduction of friction losses.

Conclusion

The analysis of the proposed hydroelectric dam at Peace River demonstrates the critical role of fluid mechanics in designing efficient renewable energy systems. Through detailed calculations, the system was found to deliver significant potential and actual power output, even after accounting for losses due to turbulence and friction. Turbulent flow, which is abundant in the penstock due to high Reynolds numbers, introduces head loss, typical for large scale systems and can be optimized through proper design and maintenance. While hydroelectric systems offer numerous environmental and operational benefits, they also pose ecological challenges that must be addressed during planning and construction. Overall, the dam design is both technically and environmentally viable as a clean energy solution for remote communities.

Recommendations

To enhance the efficiency and sustainability of this system I have come up with three recommendations; Firstly, reducing friction losses in the penstock should be a design priority, this can be achieved by using smoother internal materials, increasing the pipe diameter, and minimizing sharp bends in the flow path. Secondly, environmental considerations should guide the project implementation, including fish passage systems, sediment management, and periodic ecological impact assessments. Finally the operational efficiency can be improved through regular maintenance of the turbine and penstock to prevent wear. These recommendations will help ensure that the system operates at high efficiency while minimizing ecological disruption.

References

[1]F. M. White and H. Xue, *Fluid mechanics*, 9th ed. New York, Ny Mcgraw-Hill, 2021.

[2]E. Kjeang, *Fluid mechanics class notes and recordings*, 9th ed. Surrey, Simon Fraser University, 2025.

Appendix

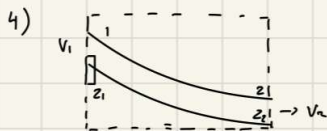
2) $\rho g H$ $H = 30\text{m}$ $\rho = 1000 \text{ kg/m}^3$ $g = 9.81 \text{ m/s}^2$

$$1000(9.81)(30) = 294.5 \text{ kPa}$$

3) $F = \cancel{P_a}^0 + \rho g h_{ca} A$ $A = h(w) = (35)(500) = 1750 \text{ m}^2$
 $h_{ca} = 35/2 = 17.5 \text{ m}$

$$F = \rho g h_{ca} A = (1000)(9.81)(17.5)(1750) = 300.43 \text{ MN}$$

P_a acts on both sides and cancels out



Assumptions

$$z_1 = 30 - 17.5 = 12.5 \text{ m}$$

Pen stock approx $\frac{1}{2}$ height of wall

$$\frac{P_1}{\rho} + \frac{1}{2} V_1^2 + g z_1 = \frac{P_2}{\rho} + \frac{1}{2} V_2^2 + g z_2 = C$$

Entrance and Exit exposed to P_{atm}

$$V_1 = \frac{Q}{A} = \frac{40}{\pi(1)^2} = 12.73 \text{ m/s}$$

$$\frac{1}{2} V_1^2 + g z_1 = \frac{1}{2} V_2^2$$

$$V_2 = \sqrt{V_1^2 + 2g z_1} = \sqrt{12.73^2 + 2(9.81)(12.5)} = 20.18 \text{ m/s}$$

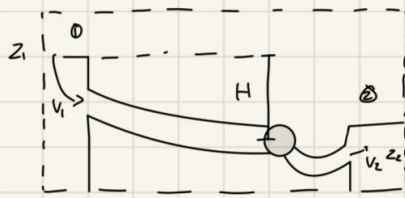
5) $(\frac{1}{2} V_1^2) + g z_1 = \frac{P_2}{\rho} + \frac{1}{2} V_2^2$ $z_1 = 12.5 - 10 = 2.5$

$$= P_2 = \rho \left(\frac{1}{2} V_1^2 + g z_1 - \frac{1}{2} V_2^2 \right) \rightarrow (1000) \left(\frac{1}{2} (12.73)^2 + (9.81)(2.5) - \frac{1}{2} (20.18)^2 \right)$$

$$P_2 = -98.07 \text{ kPa}$$

Figure 1, Questions 2 - 5

6)



Assumptions

$$v_1 = v_2 = 0$$

$$z_1 - z_2 = H$$

Steady Flow

No friction head loss

$$\frac{P_1}{\rho} + \frac{v_1^2}{2g} + z_1 = \frac{P_2}{\rho} + \frac{v_2^2}{2g} + z_2 + h_f - h_p + h_t \quad P_1 = P_2 = P_{atm}$$

$$z_1 = z_2 + h_t \rightarrow H = h_t = 30m$$

$$\text{Shaft work rate } \dot{W}_s = \dot{m} w_s \rightarrow (\rho Q) (g h_t) = (1000)(40)(9.81)(30)$$

$$\dot{W}_s = 11.77 \text{ MW}$$

$$7) W_{electric} = \eta_{gen} \dot{W}_s \quad \eta_{gen} = 0.85 \quad \dot{W}_s = 11.77 \text{ MW}$$

$$W_{electric} = (0.85)(11.77) = 10 \text{ MW}$$

$$8) R_e = \frac{Vd}{\nu} \quad \nu = 1.002 \times 10^{-3} \text{ Pa}\cdot\text{s} \quad d = 2m \quad V = \frac{40}{\pi(1)^2} = 12.73 \text{ m/s}$$

Turbulent Flow

$$R_e = \frac{12.73(2)}{1.002 \times 10^{-3}} = 25409 \quad R_e > 10^4$$

9) Assume Rough Concrete Table 6.1

$$\epsilon = 2.00 \text{ mm} \pm 50\%$$

$$V = 12.73 \text{ m/s} \quad L = 60m$$

$$\frac{\epsilon}{d} = \frac{2}{2000} = 0.01$$

$$\text{Moody chart, } R_e = 25000, \epsilon/d = 0.001 \quad f \approx 0.027$$

$$h_f = f \frac{L}{d} \frac{V^2}{2g} = (0.027) \left(\frac{60}{2} \right) \left(\frac{12.73^2}{2(9.81)} \right)$$

$$h_f = 6.69 \text{ m}$$

$$10) z_1 = z_2 + h_f + h_t \quad H = 30m \quad h_f = 6.69m \quad z_1 - z_2 = H$$

$$2 \text{ of } 3 \quad h_t = H - h_f = 30 - 6.69$$

$$h_t = 23.31 \text{ m}$$

Figure 1, Questions 6 - 10